Integral Imprecise Probability Metrics

Krikamol Muandet

CISPA Helmholtz Center for Information Security
Saarbrücken, Germany

Joint work with **Siu Lun Chau** (NTU) and **Michele Caprio** (Manchester)

March 28, 2026

Table of Contents

1 Motivation

2 Imprecise Probabilities

3 Integral Imprecise Probability Metrics

4 Application: Selective Classification

Integral Probability Metrics

Integral Probability Metrics (Müller, 1997)

Given a set of continuous bounded real-valued measurable functions $\mathcal{F} \subseteq C_b(\mathcal{X})$ and probability measures $\mathbb{P}, \mathbb{Q} \in \mathcal{P}(\mathcal{X})$, the integral probability metric (IPM) associated to $\mathcal{F}$ is

$$\mathrm{IPM}_{\mathcal{F}}(\mathbb{P}, \mathbb{Q}) = \sup_{f \in \mathcal{F}} \left| \int f \, d\mathbb{P} - \int f \, d\mathbb{Q} \right|.$$

- Popular examples include the total variation distance, Wasserstein distance, and maximum mean discrepancy (MMD):

$$\mathrm{IPM}_{\mathcal{H}_k}(\mathbb{P}, \mathbb{Q}) = \|\mu_{\mathbb{P}} - \mu_{\mathbb{Q}}\|^2_{\mathcal{H}_k}.$$

- Two-sample testing, generative modeling, causal inference, domain adaptation, fairness and algorithmic auditing, etc.

Integral Imprecise Probability Metrics

Our Contributions

- We extend IPM to **imprecise probabilities**:

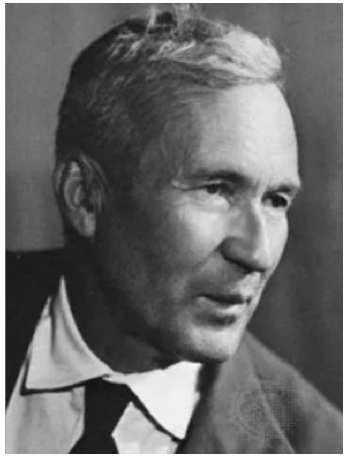
$$\mathrm{IPM}_{\mathcal{F}}(\mathbb{P},\mathbb{Q}) = \sup_{f\in\mathcal{F}} \left| \int f \, d\mathbb{P} - \int f \, d\mathbb{Q} \right|$$

$$\Downarrow$$

$$\mathrm{IIPM}_{\mathcal{F}}(\nu,\mu) = \sup_{f\in\mathcal{F}} \left| \oint f \, d\nu - \oint f \, d\mu \right|$$

- We establish the **theoretical properties** of IIPM, including its metric properties and weak convergence under varying choice of function classes $\mathcal{F}$ and capacities ν, μ.
- We apply IIPM to **epistemic** uncertainty quantification.

(Additive) Probability Theory



Probability (Kolmogorov, 1933)

A probability measure $\mathbb{P} \in \mathcal{P}(\mathcal{X})$ on a measurable space $(\mathcal{X}, \Sigma_{\mathcal{X}})$ is a set function $\mathbb{P} : \Sigma_{\mathcal{X}} \to [0, 1]$ such that

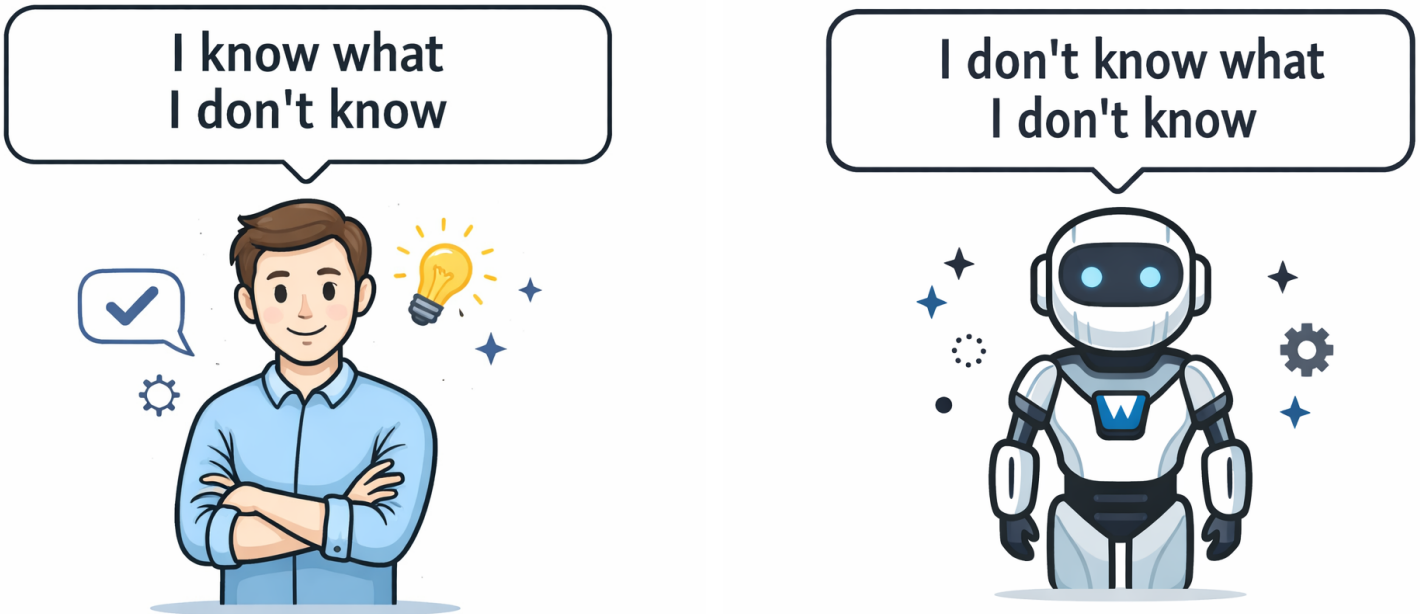
1. $\mathbb{P}(A) \geq 0$, for all event $A \in \Sigma_{\mathcal{X}}$
2. $\mathbb{P}(\mathcal{X}) = 1$,
3. for any sequence of disjoint sets $\{A_i\}_{i \geq 1}$ $\mathbb{P}(\bigcup_{i \geq 1} A_i) = \sum_{i \geq 1} \mathbb{P}(A_i)$

Limitations of Standard Probability

1. Standard probability (e.g. real-valued random variables) lack formal framework to **handle imprecise observations**:
   - Set-valued observations (random set, ontic v.s. ontological), missing observations, qualitative data, probability bounds
2. Probability fail to distinguish between **indifference** and **ignorance**.
   - $\mathbb{P}(\{\text{Coin flip} = \text{head}\}) = 0.5$. Precise equal likelihood belief or pick 0.5 because you don't know anything?
3. Probability cannot encode **absolute strength of belief**.
   - If I do not know anything about event A, I want to encode $\mathbb{P}(A) = 0$, but that implies I know everything about A^c since now $\mathbb{P}(A^c) = 1$.[1]
4. Probability cannot represent **ignorance** anyway...
   - Uniform distribution is not closed under reparametrisation.

[1] therefore probability shouldn't be used as confidence measure but people use it regardlessly..

Knowledge and Ignorance



"*Awareness of ignorance is the beginning of wisdom*"

– **Socrates**

Table of Contents

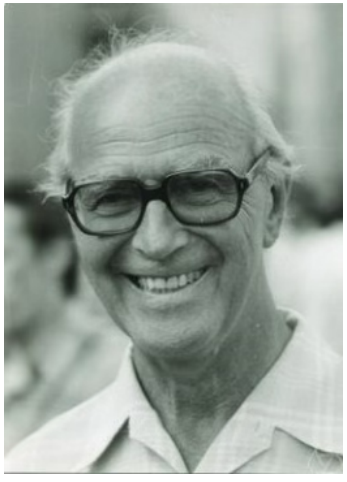
1 Motivation

2 Imprecise Probabilities

3 Integral Imprecise Probability Metrics

4 Application: Selective Classification

Capacities



Capacities (Choquet, 1954)

A capacity $\nu \in \mathcal{V}(\mathcal{X})$ on a measurable space $(\mathcal{X}, \Sigma_{\mathcal{X}})$ is a set function $\nu : \Sigma_{\mathcal{X}} \to [0, 1]$ such that

- $\nu(\emptyset) = 0, \quad \nu(\mathcal{X}) = 1$
- for any $A, B \in \Sigma_{\mathcal{X}}$, $A \subseteq B \implies \nu(A) \leq \nu(B)$.

Choquet Capacities

n-monotone Choquet capacity

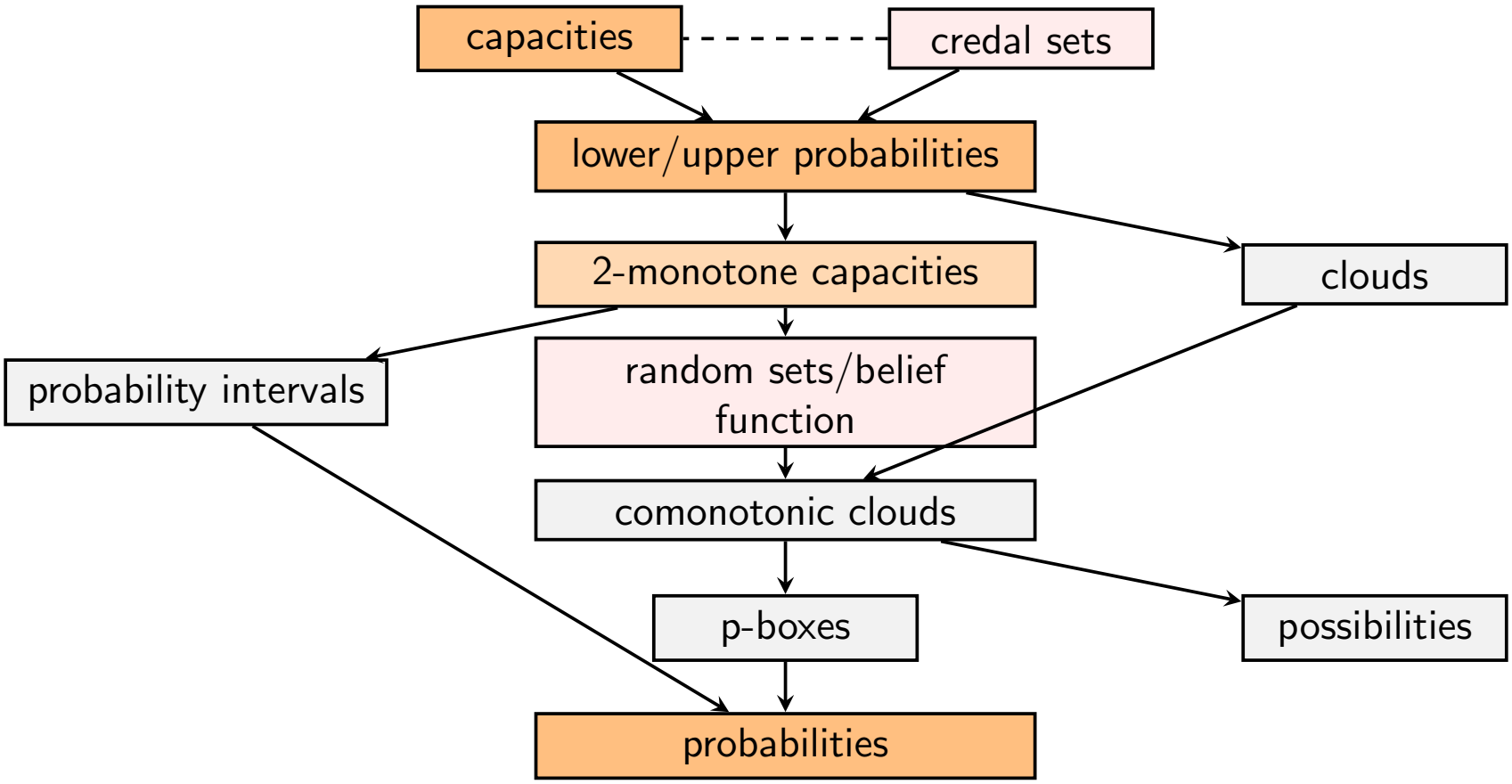
A set function $\nu(\cdot)$ is an n-**monotone Choquet capacity** if

- $\nu(\emptyset) = 0, \nu(\mathcal{X}) = 1$
- For every integer k with $1 \leq k \leq n$, and for every collection of sets $A_1, \ldots, A_k$, the following inclusion-exclusion inequality holds:

$$\sum_{\emptyset \neq I \subseteq \{1,\ldots,k\}} (-1)^{|I|+1} \nu\left(\bigcup_{i \in I} A_i\right) \geq \nu\left(\bigcap_{i=1}^{k} A_i\right).$$

- Any lower probability is 1-monotone and 2-monotone lower probabilities are 2-monotone capacities.
- If a lower probability is n-monotone for all n, then it is called **infinite monotone** or **belief function**.

Generalisation of (Imprecise) Probabilities



Properties of Capacities

For some capacity ν:

- $\nu(A) \leftarrow$ credence to the truthfulness of event A.
- $\nu(A^c) \leftarrow$ credence to the truthfulness of complementary event A^c
- In general, $\nu(A) \neq 1 - \nu(A^c)$! (unlike precise probability)

Conjugate capacity

Given a capacity ν, the conjugate capacity $\nu^\star : \Sigma_\mathcal{X} \to [0, 1]$ is given by

$$\nu^\star(A) = 1 - \nu(A^c)$$

Credence for A is now given by the non-truthfulness of event A^c.

Example (Lower and upper probabilities)

$$\sup_{\mathbb{P} \in C} P(A) = \overline{P}(A) = 1 - \underline{P}(A^c) = 1 - \inf_{P \in C} P(A^c)$$

Choquet Integration

Choquet Integrals (Choquet, 1954)

Given a capacity $\nu \in \mathcal{V}(\mathcal{X})$ and a real-valued function f on $\mathcal{X}$, the Choquet integral of f with respect to ν is

$$\oint f \, d\nu = \int_0^\infty \nu(\{x : \max(0, f(x)) \geq t\})dt - \int_0^\infty \nu(\{x : -\min(0, f(x)) \geq t\})dt$$

provided the difference is well defined. For bounded f, we have

$$\oint f \, d\nu = \underline{f} + \int_{\underline{f}}^{\overline{f}} \nu(\{x : f(x) \geq t\})dt$$

where $\underline{f} = \inf_x f(x)$ and $\overline{f} = \sup_x f(x)$.

Table of Contents

1 Motivation

2 Imprecise Probabilities

3 Integral Imprecise Probability Metrics

4 Application: Selective Classification

Integral Imprecise Probability Metric (IIPM)

IIPM (Chau et al., 2025)

For function class $\mathcal{F} \subseteq C_b(\mathcal{X})$ and capacities $\nu, \mu \in \mathcal{V}(\mathcal{X})$, the integral imprecise probability metric associated with $\mathcal{F}$ between ν, μ is

$$\mathrm{IIPM}_{\mathcal{F}}(\nu, \mu) = \sup_{f \in \mathcal{F}} \left| \oint f \, d\nu - \oint f \, d\mu \right|$$

Theoretical Results

- When $\mathcal{F}$ is dense in $C_b(\mathcal{X})$, IIPM is a metric on $\mathcal{V}(\mathcal{X})$.
- When ν, μ is an additive probability, IIPM recovers IPM.
- For $\mathcal{F} \subseteq C_b(\mathcal{X})$ that are dense, $\mathrm{IIPM}_{\mathcal{F}}$ metrises the (Choquet) weak convergence of $\mathcal{V}(\mathcal{X})$.

Maximum Mean Imprecision (MMI)

The Maximum Mean Imprecision (MMI)

The **Maximum Mean Imprecision** with respect to $\mathcal{F} \subseteq C_b(\mathcal{X})$ is

$$\mathrm{MMI}_{\mathcal{F}}(\underline{\mathbb{P}}) = \mathrm{IIPM}_{\mathcal{F}}(\underline{\mathbb{P}}, \overline{\mathbb{P}}) = \sup_{f \in \mathcal{F}} \left\{ \oint f \, d\overline{\mathbb{P}} - \oint f \, d\underline{\mathbb{P}} \right\}$$

Properties

- Unlike MMD, the MMI quantifies the largest "mean" difference between the lower probability and its conjugate.
- $\mathrm{MMI}_{\mathcal{F}}(\mathbb{P}) = 0$ and increases with the imprecision of $\underline{\mathbb{P}}$.
- Perfect as a measure of **epistemic** uncertainty.

Measures of Epistemic Uncertainty

Desirable Properties

Let $\mathbb{U} : \underline{\mathbb{P}}(\mathcal{X}) \to \mathbb{R}$ be a credal uncertainty measure:

1. Non-negative and boundedness
2. Continuity: $\mathbb{U}$ is a continuos functional
3. Monotonicity: $\mathbb{U}(\underline{\mathbb{P}}) \geq \mathbb{U}(\underline{\mathbb{Q}})$ if $\underline{\mathbb{P}}(A) \leq \underline{\mathbb{Q}}(A)$ for all $A \in \Sigma_{\mathcal{X}}$.
4. Probability consistency: $\mathbb{U}(\mathbb{P}) = 0$ for all precise $\mathbb{P} \in \mathcal{P}(\mathcal{X})$.
5. Sub-additivity $\mathbb{U}(\underline{P}_{\mathcal{X}\times\mathcal{Y}}) \leq \mathbb{U}(\underline{P}_{\mathcal{X}}) + \mathbb{U}(\underline{P}_{\mathcal{Y}})$
6. Additivity (5) becomes equality when $\underline{P}_{\mathcal{X}}$ and $\underline{P}_{\mathcal{Y}}$ are **independent**.

MMI satisfy 1–4 for any $\mathcal{F}$, and the rest when $\underline{P}$ satisfies 2-monotonicity.

Table of Contents

1 Motivation

2 Imprecise Probabilities

3 Integral Imprecise Probability Metrics

4 Application: Selective Classification

Selective Classification

For each data point $i \in 1, \ldots, n_{\text{test}}$,

$$x_i \mapsto \Big(\underbrace{\hat{y}_i}_{\text{prediction}} \;,\; \underbrace{\text{EU}(x_i)}_{\text{predictive EU}} \Big)$$

1. Sort $x_1, \ldots, x_{n_{\text{test}}}$ in ascending order based on their estimated predictive confidence (inverse EU).
2. Remove the top $p\%$ of least confident inputs, predict the rest, record performance.
3. Plot performance against threshold $p\%$

Uncertainty Quantification

- Entropy Difference (Abellán et al., 2006)

$$\mathbb{U}_{\text{E-diff}}(\underline{\mathbb{P}}) = H_{\max} - H_{\min}, \quad H(\mathbb{P}) = -\sum_{i=1}^{K} p_i \log p_i$$

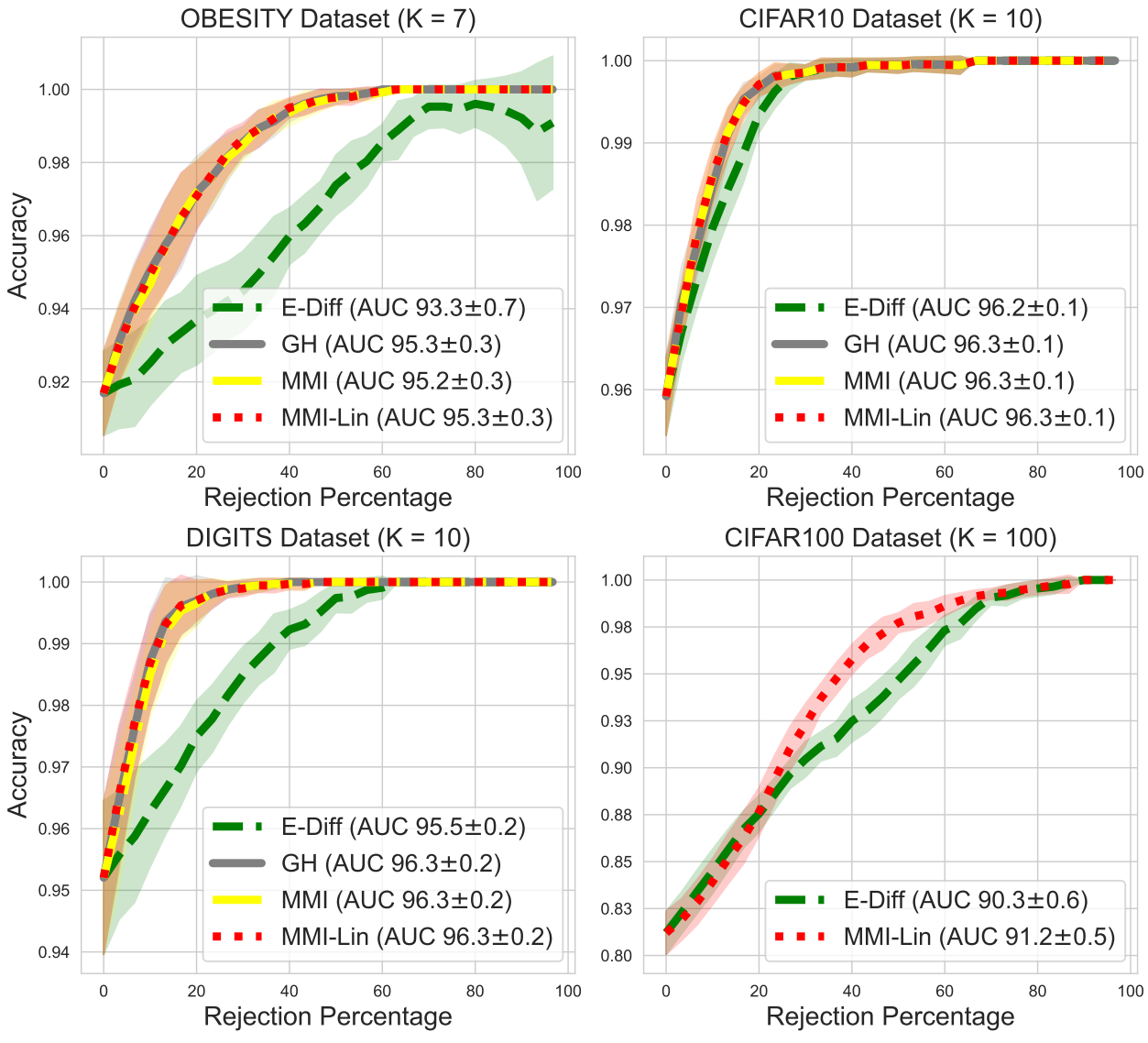
- Generalised Hartley (GH) (Abellán and Gómez, 2006)

$$\mathbb{U}_{\text{GH}}(\underline{\mathbb{P}}) = \sum_{A \subseteq \mathcal{X}} m_{\underline{\mathbb{P}}}(A) \log(|A|)$$

- $\text{MMI}_{\mathcal{H}_{TV}}(\underline{\mathbb{P}}) = \sup_{A \subseteq \mathcal{Y}} \left| \overline{\mathbb{P}}(A) - \underline{\mathbb{P}}(A) \right|$
- MMI-Lin:

$$\text{MMI}_{\mathcal{H}_{TV}}(\underline{\mathbb{P}}) \leq 1 - \sum_{y \in \mathcal{Y}} \underline{\mathbb{P}}(\{y\})$$

Experiment Results



Further Information

Integral Imprecise Probability Metrics

Siu Lun Chau[1], **Michele Caprio**[2,3], and **Krikamol Muandet**[4]

[1]College of Computing & Data Science, Nanyang Technological University, Singapore
[2]Department of Computer Science, The University of Manchester, United Kingdom
[3]Manchester Centre for AI Fundamentals, Manchester, United Kingdom
[4]Rational Intelligence Lab, CISPA Helmholtz Center for Information Security, Germany

Abstract

Quantifying differences between probability distributions is fundamental to statistics and machine learning, primarily for comparing statistical uncertainty. In contrast, epistemic uncertainty—due to incomplete knowledge—requires richer representations than those offered by classical probability. Imprecise probability (IP) theory offers such models, capturing ambiguity and partial belief. This has

Neural Information Processing Systems (NeurIPS) 2025

Thank You

For more information, please visit https://ri-lab.org/



CISPA Helmholtz Center for Information Security, Saarbrücken, Germany

References I

Joaquín Abellán and Manuel Gómez. Measures of divergence on credal sets. *Fuzzy Sets and Systems*, 157(11):1514–1531, June 2006. ISSN 01650114. doi: 10.1016/j.fss.2005.11.021.

Joaquín Abellán, George Jiří Klir, and Serafín Moral. Disaggregated total uncertainty measure for credal sets. *International Journal of General Systems*, 1(35):29–44, 2006.

Siu Lun Chau, Michele Caprio, and Krikamol Muandet. Integral imprecise probability metrics. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025. URL https://openreview.net/forum?id=KM2XzHq2Rm.

Gustave Choquet. Theory of capacities. *Annales de l'Institut Fourier*, 5: 131–295, 1954.

Kolmogorov. Sulla determinazione empirica di una legge didistribuzione. *Giorn Dell'inst Ital Degli Att*, 4:89–91, 1933.

Alfred Müller. Integral probability metrics and their generating classes of functions. *Advances in applied probability*, 29(2):429–443, 1997.